

AWS + Docker + Kubernetes Project Commands Guide

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This document contains all major commands used throughout the project including Docker setup, AWS ECR integration, EC2 configuration, Minikube setup, Kubernetes deployment, scaling, debugging, and cleanup.

Phase 1 — Docker Build & Run

Command	Purpose
<code>docker build -t node-k8s-app -f docker/Dockerfile .</code>	Builds Docker image from Dockerfile.
<code>docker images</code>	Lists all Docker images available locally.
<code>docker run -d -p 3000:3000 --name node-container node-k8s-app</code>	Runs the Docker container in detached mode.
<code>docker ps</code>	Displays running Docker containers.
<code>docker ps -a</code>	Displays all containers including stopped ones.
<code>docker rm node-container</code>	Removes an existing container.

Phase 2 — AWS CLI & ECR Setup

Command	Purpose
<code>aws configure</code>	Configures AWS CLI credentials and region.
<code>aws sts get-caller-identity</code>	Verifies current AWS account login.
<code>aws ecr get-login-password --region us-east-1 docker login --username AWS --password-stdin 228235827513.dkr.ecr.us-east-1.amazonaws.com</code>	Authenticates Docker with Amazon ECR.
<code>docker tag node-k8s-app:latest 228235827513.dkr.ecr.us-east-1.amazonaws.com/node-k8s-app:latest</code>	Tags Docker image for ECR repository.
<code>docker push 228235827513.dkr.ecr.us-east-1.amazonaws.com/node-k8s-app:latest</code>	Pushes Docker image to Amazon ECR.

Phase 3 — EC2 SSH Connection

Command	Purpose
<code>chmod 400 minikube-key.pem</code>	Sets secure permissions for SSH key.
<code>ssh -i "minikube-key.pem" ubuntu@PUBLIC_IP</code>	Connects to EC2 instance using SSH.

Phase 4 — Docker Installation on EC2

Command	Purpose
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<code>sudo apt update && sudo apt upgrade -y</code>	Updates Ubuntu packages.
<code>sudo apt install docker.io -y</code>	Installs Docker engine.
<code>sudo systemctl start docker</code>	Starts Docker service.
<code>sudo systemctl enable docker</code>	Enables Docker on boot.
<code>sudo usermod -aG docker ubuntu</code>	Adds ubuntu user to Docker group.
<code>newgrp docker</code>	Applies Docker group changes.
<code>docker --version</code>	Checks installed Docker version.

Phase 5 — kubectl & Minikube Setup

Command	Purpose
<code>curl -LO "https://dl.k8s.io/release/\$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"</code>	Downloads kubectl.
<code>chmod +x kubectl</code>	Makes kubectl executable.
<code>sudo mv kubectl /usr/local/bin/</code>	Moves kubectl to system path.
<code>kubectl version --client</code>	Checks kubectl version.
<code>curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64</code>	Downloads Minikube binary.
<code>sudo install minikube-linux-amd64 /usr/local/bin/minikube</code>	Installs Minikube.
<code>minikube version</code>	Checks Minikube version.
<code>minikube start --driver=docker --memory=1800mb --cpus=2</code>	Starts Minikube Kubernetes cluster.
<code>kubectl get nodes</code>	Displays Kubernetes nodes.

Phase 6 — AWS CLI on EC2

Command	Purpose
<code>curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"</code>	Downloads AWS CLI installer.
<code>sudo apt install unzip -y</code>	Installs unzip package.
<code>unzip awscliv2.zip</code>	Extracts AWS CLI installer.
<code>sudo ./aws/install</code>	Installs AWS CLI.
<code>aws --version</code>	Checks AWS CLI version.
<code>docker pull 228235827513.dkr.ecr.us-east-1.amazonaws.com/node-k8s-app:latest</code>	Pulls Docker image from ECR.

Phase 7 — Kubernetes Deployment

Command	Purpose
<code>nano deployment.yaml</code>	Creates Kubernetes deployment configuration.
<code>nano service.yaml</code>	Creates Kubernetes service configuration.

kubectl apply -f deployment.yaml	Deploys application to Kubernetes.
kubectl apply -f service.yaml	Creates NodePort service.
kubectl get pods	Lists running Kubernetes pods.
kubectl get services	Lists Kubernetes services.
kubectl describe pod POD_NAME	Displays detailed pod information.
kubectl logs deployment/node-k8s-deployment	Shows application logs.

Phase 8 — ECR Authentication Fix

Command	Purpose
kubectl create secret docker-registry ecr-secret --docker-server=228235827513.dkr.ecr.us-east-1.amazonaws.com --docker-username=AWS --docker-password=\$(cat /dev/urandom tr -dc 'a-z0-9' fold -n 40 tr -d '\n' fold -n 1 cat)	Creates Kubernetes secret for ECR authentication.
minikube image load 228235827513.dkr.ecr.us-east-1.amazonaws.com/node-k8s-app:latest	Loads Docker image into Minikube runtime.

Phase 9 — Public Access Fix

Command	Purpose
sudo apt install socat -y	Installs socat for port forwarding.
sudo socat TCP-LISTEN:30007,fork TCP:\$(minikube ip):30007	Forwards EC2 public traffic to Minikube NodePort.

Phase 10 — Scaling & Monitoring

Command	Purpose
kubectl scale deployment node-k8s-deployment --replicas=2	Scales deployment to two replicas.
kubectl get pods -o wide	Displays pod details with node and IP information.
kubectl describe service node-k8s-service	Displays service details.

Phase 11 — Cleanup Commands

Command	Purpose
kubectl delete service node-k8s-service	Deletes Kubernetes service.
kubectl delete deployment node-k8s-deployment	Deletes deployment.
minikube stop	Stops Minikube cluster.
minikube delete	Deletes Minikube cluster completely.
exit	Exits EC2 SSH session.